

$$1. \quad f(x) \rightarrow f(x, y) = 40x^2 - 2xy + y^2 - 3x + 2y + 7$$

$$\frac{df}{dx} = 8x - 2y - 3$$

$$\frac{df}{dy} = -2x + 2y + 2$$

$$x_0 = (2, 2) \text{ at } \alpha = 0.15$$

$$\frac{df}{dx} \Rightarrow 8(2) - 2(2) - 3 = 9$$

$$\frac{df}{dy} \Rightarrow -2(2) + 2(2) + 2 = 2$$

$$\partial C_1 = \partial C_0 - \alpha \nabla f(\partial C_0) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - 0.15 \begin{pmatrix} 9 \\ 2 \end{pmatrix} = \begin{pmatrix} 0.65 \\ 1.7 \end{pmatrix}$$

$$\text{sub } \partial C_1, \quad \frac{df}{dx} = -1.2, \quad \frac{df}{dy} = 4.1$$

$$\partial C_2 = \begin{pmatrix} 0.65 \\ 1.7 \end{pmatrix} - 0.15 \begin{pmatrix} -1.2 \\ 4.1 \end{pmatrix} = \begin{pmatrix} 0.83 \\ 1.085 \end{pmatrix}$$

- (ii) GD uses entire dataset each calculation, stochastic GD samples a random index per iteration

$$2. (i) \bar{x} = 1.73 +$$

$$\bar{x} = \frac{1.73 + 1.87 + 1.9 + 2.66 + 3.43 + 4.22}{6}$$

$$= 2.6350$$

$$\bar{y} = \frac{4.61 + 4.98 + 4.39 + 6.59 + 7.02 + 9.65}{6}$$

$$\bar{y} = 6.3067$$

$$\hat{\beta}_1 = \frac{\sum x_i y_i - n(\bar{x})(\bar{y})}{\sum x_i^2 - n(\bar{x})^2} = \frac{110.0179 - 99.7305}{46.7487 - 41.6414} = 2.0143 \rightarrow 2.56$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.99906.9692$$

$$\therefore \hat{\beta}_0 = 0.9692, \hat{\beta}_1 = 2.0256$$

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3 (i) ~~σc_i~~ x_i

$$z = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$p(y_i = 1 | x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})}}$$

$$p(y_i = 0 | x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})}}$$

(ii) $\beta_0^* = 1.8183, \dots$

$$z = 1.8183 + (-0.0665)(70) + 0$$
$$= -2.8367$$

$$p(y=1|x) = \frac{1}{1 + e^{-z}} = 0.9446$$
$$= 0.05537$$
$$= 0.0554$$

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3 (iii) since $f(x) = 0.0554 < 0.5$
 $= r$, $\hat{y} = 0$

$\Rightarrow$ Death